

Supplementary Material

for “AdaScale: Diagnosing and Adapting Multi-Scale Fusion for Training-Free Zero-Shot Recognition”

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1 Implementation Details

Multi-scale feature extraction. For each dataset we extract 10 per-scale feature sets following the reference LG-CLIP pipeline [1]: each test image is cropped by `RandomResizedCrop(224, scale = (s/10, s/10))` with BICUBIC interpolation and CLIP normalization, then encoded by the frozen CLIP image encoder into an L2-normalized feature. Per-scale features are stored as HDF5 files; text prototypes use the template “A photo of a {class}.”. Script: `third_party/LG-CLIP/vanilla_clip_ms.py`; all evaluation is training-free and gradient-free.

CUB-200-2011. Official train/test split (5,794 test images, 200 classes); class names cleaned as in the reference pipeline. Only text prototypes are evaluated on CUB (synthetic prototypes require offline per-class image generation and are covered by the other three datasets).

Frozen hyperparameters. Early-exit $r=4$, $\delta=0.65$, $\varepsilon=0.01$; selector $\theta=0.01$, $n=500$ unlabeled calibration samples drawn from the target test pool (labels never used); soft-fusion temperature $\tau_s=0.02$. **None of these is tuned on test labels**; configurations were fixed from the calibration-side search reported in Sec. 3 and Sec. 5.

2 AdaScale Algorithm

3 Selector Sensitivity Analysis

The Level-1 decision uses a single scalar on unlabeled data. Its robustness was verified as follows.

- **Calibration sub-sampling:** 200 uniform sub-samples of size $n=500$ per setting produced the same SINGLE/FUSED decision as the full set in 98–100% of runs (100% at $n=1000$).
- **Threshold tolerance:** the decision is invariant for $\theta \in [0.5\%, 2.3\%]$; the observed ΔH gap between regimes is +2.20pp (minimum SINGLE-side value +2.66% at EuroSAT/gen, maximum FUSED-side value +0.46% at Flowers/gen).
- **Cross-backbone:** 11/11 correct decisions across CLIP ViT-B/32 and ViT-B/16 (Table 1 and main Table III).
- **Failure mode is safe:** a wrong FUSED decision would forfeit compute savings but keep full-fusion accuracy.

Algorithm 1 AdaScale: hierarchical training-free controller

Require: test image x ; unlabeled calibration set $\mathcal{U}$ ($n=500$); frozen config $(\theta, r, \delta, \varepsilon, \tau_s)$; prototypes P .

```
1: Compute fused and largest-scale predictions on  $\mathcal{U}$ ;  $\Delta H \leftarrow \frac{1}{n} \sum_i \frac{H(p_i^{\text{fused}}) - H(p_i^{\text{single}})}{\log C}$   
2: if  $\Delta H > \theta$  then ▷ Level 1: target-domain fusion diagnosis  
3:   return largest-scale prediction ▷ SINGLE: 1 forward  
4: else ▷ FUSED: Level 2 loop  
5:    $\mathcal{V} \leftarrow \emptyset$ ; prefix  $\leftarrow 0$   
6:   for  $t = 1, \dots, 10$  do  
7:      $f_t \leftarrow \text{Encode}(\text{RRC}(x, t/10))$ ;  $\mathcal{V} \leftarrow \mathcal{V} \cup \{t\}$   
8:      $p^{(t)} \leftarrow \text{softmax}(\text{prefix})$  ▷ prefix fusion  
9:     if  $t \geq r$  and  $\arg \max p^{(t-j)}$  equal for  $j=0..r-1$  and  $m_t \geq \delta$  and  $D_{\text{JS}}(p^{(t)} \| p^{(t-1)}) \leq \varepsilon$  then  
10:      break ▷ sample-wise exit  
11:   return  $\arg \max_c \sum_{s \in \mathcal{V}} s \cdot \text{softmax}(f_s^\top P / \tau_s)_c$  ▷ probability-level soft fusion
```

Prefix dynamics by execution group: early exits stabilize fast; full runs stay low-margin (wrong ones lowest)

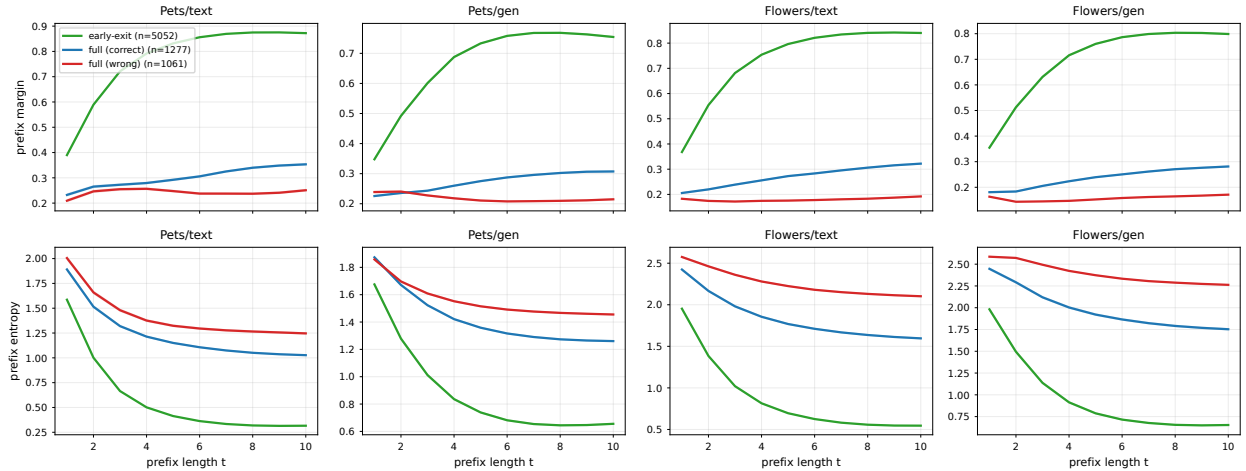


Figure 1: Prefix margin (top) and entropy (bottom) trajectories by execution group.

4 Additional Diagnostics

Prefix dynamics. Figure 1 shows prefix decision margin and entropy trajectories grouped by execution outcome on the four FUSED settings (ViT-B/32). Early-exit samples stabilize fast (margin ≈ 0.9 after 3–4 scales); full-run correct samples remain at intermediate margin (≈ 0.3); full-run wrong samples have the lowest margin (≈ 0.2) and the highest entropy throughout — the stopping rule separates exactly the regimes it is designed to separate.

Symptom vs. signal: quadrant analysis and the drag statistic. Figure 3 reproduces the quadrant analysis of the main paper (main Fig. 4). The $s_1 \times s_{10}$ correctness quadrants and the NQ-rescue rate. The window between “the fusion operator is beneficial” and “harmful” domains is directly visible: rescue rates are 92–97% on Pets/Flowers/CUB but only 76–85% on EuroSAT. The label-free “drag” statistic — over samples where s_1 and s_{10} disagree, the fraction where the fused prediction follows s_1 — shows the same separation in the direction of displacement (18.4% for EuroSAT/gen vs. $\leq 6.8\%$ elsewhere), but its regime separation (+1.01pp) is weaker than the entropy gap (+2.20pp), motivating the entropy-based selector (Sec. 3).

Attention statistics (negative result, reported for completeness). A per-sample quantitative study of CLIP attention rollout (600 sampled EuroSAT images $\times$ 10 scales) found the normalized attention entropy to be nearly constant (all 12 layers > 0.91 ; 0.998 after rollout) and uncorrelated with correctness or scale.

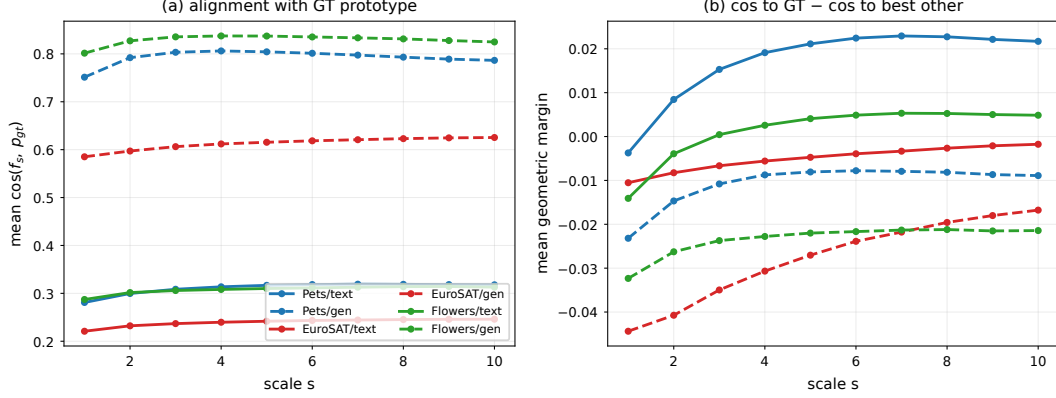


Figure 2: Feature geometry by scale.

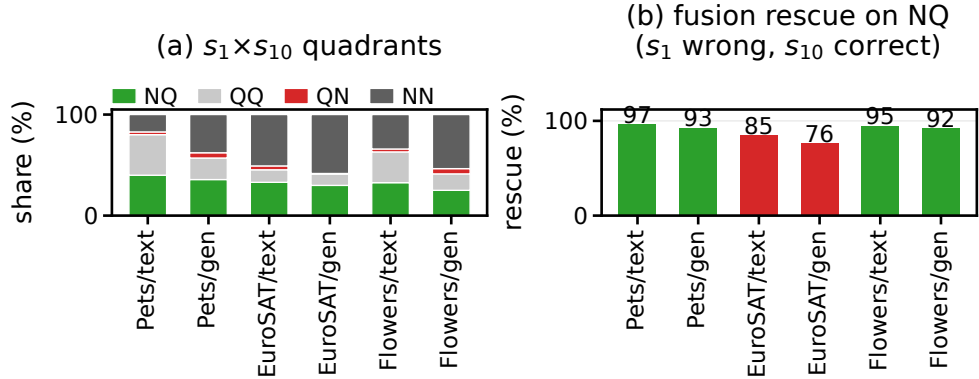


Figure 3: Microscopic view of the domain-level fusion failure (reproduced from the main paper, Fig. 4).

CLIP attention is diffuse, so the qualitative overlays in the main paper (main Fig. 3) are per-crop min–max normalized for display; they are used to illustrate the local-texture-to-global-structure transition, not as a quantitative diagnostic. This negative result is why AdaScale uses prediction-space statistics (entropy, margin, JS) rather than attention statistics.

Feature geometry. Figure 2 reports the mean cosine similarity to the ground-truth prototype and the geometric margin ($\cos(f_s, p_{gt}) - \max_{c \neq gt} \cos(f_s, p_c)$) per scale. Alignment increases with scale for all settings; synthetic prototypes plateau at a lower geometric margin, consistent with their different calibration.

5 Additional Results

ViT-B/16 full results. Table 1 reports all ViT-B/16 experiments. The selector is correct on all four settings; AdaScale matches or exceeds Full-10 except 0.04 points on Pets/text (n.s.), with 15–90% fewer scale evaluations. Note the EuroSAT single-scale advantage reproduces exactly on the larger backbone (42.74 vs. 40.47 fused).

Runtime. Table 2 (reproduced) reports wall-clock timings on a single RTX 4080 Super, batch size 32, crops shared bit-exactly between methods.

Soft-fusion (v3) sensitivity and gain decomposition. Across the eight fused-mode settings, probability-level soft fusion improves accuracy by +0.15 on average (all non-negative; significant on Flowers-102/gen $p=0.002$ and Pets/gen at ViT-B/16 $p<0.01$, n.s. elsewhere). The gain is fully attributable to hard samples: samples that

Table 1: ViT-B/16 results. ΔH : calibration entropy gap (threshold $\theta=1\%$). v3: probability-level soft fusion.

Setting	Mode	Full-10	Single	v2	v3	#Scales
Pets/text	FUSED	86.14	84.53	86.08	86.10	6.42
Pets/gen	FUSED	64.17	61.96	64.15	64.49	8.50
EuroSAT/text	SINGLE	40.47	42.74	42.74	—	1.00
Flowers/text	FUSED	68.10	66.37	68.12	68.23	7.53

Table 2: Wall-clock inference (batch 32, RTX 4080 Super). PET: $N=7390$; EuroSAT: $N=27000$; Single mode evaluates only the largest scale.

Method	Time (s)	Ips.	Acc (%)	Avg. scales
Full-10 (PET)	290	25.4	82.50	10.00
Fixed-5 (PET)	127	58.3	81.15	5.00
Fixed-7 (PET)	189	39.0	82.75	7.00
AdaScale (PET)	202	36.6	82.37	6.64
Full-10 (EURO; 27k)	397	68.0	42.35	10.00
Single (EURO; 27k)	45	595.8	45.34	1.00

run to completion gain $+0.27$ – $+0.56$, while early-exit samples change by exactly 0.00 (the decision layer is frozen, so runtime is unchanged). Temperature sensitivity: $\tau_s \in \{0.01, 0.02, 0.05\}$ are all non-negative, with 0.02 most stable across settings.

Where fusion rescue degrades (mechanism). On EuroSAT, among samples where s_1 is wrong but s_{10} is correct (NQ, 33–38% of the dataset), the fused predictor recovers only 84.6%/76.2% (text/gen, ViT-B/32) and 85.1% (text, ViT-B/16), versus 92–97% on all other datasets. This is the microscopic counterpart of the domain-level failure and is reported in main Fig. 4.

References

- [1] D. Shi, D. Fu, Y. Liu, and J. Wang, “Can synthetic images serve as effective and efficient class prototypes?,” in *ICASSP*, 2026.